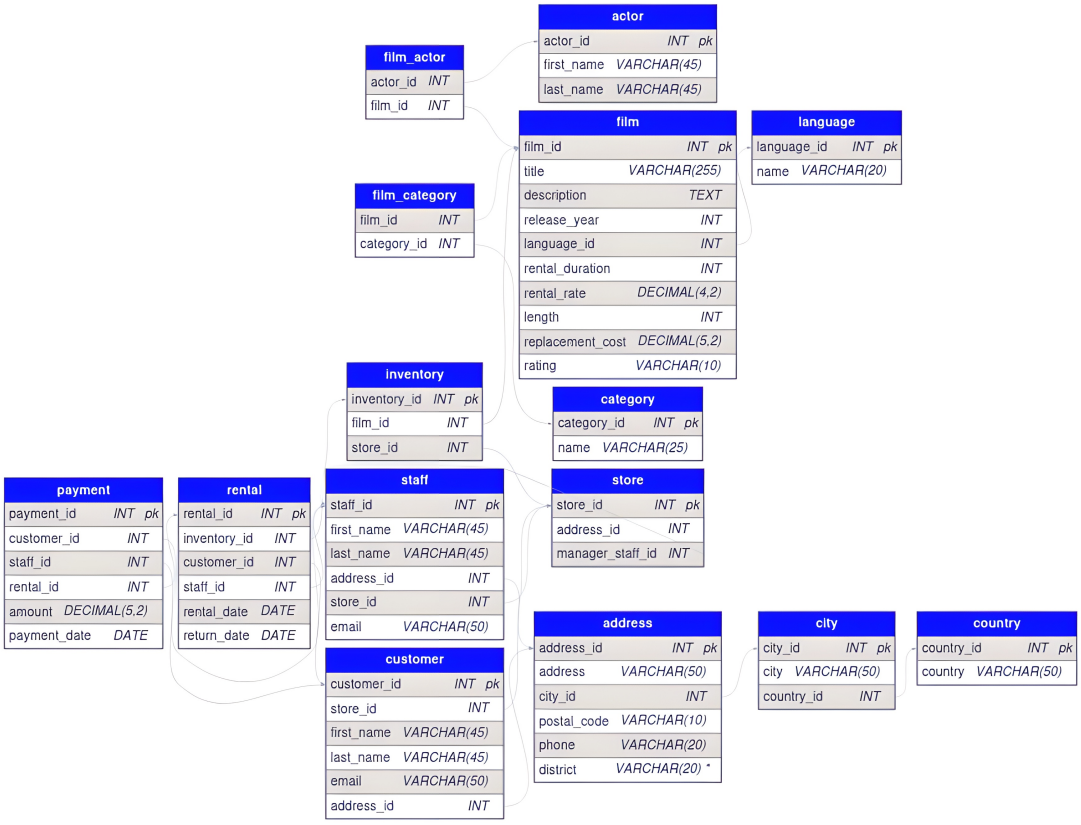
Lab Mid

CSE 4308

DATABASE MANAGEMENT SYSTEMS LAB

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DBMS Lab Mid



1. List all addresses where the postal code is either missing or empty.

```sql
SELECT address_id, address, city_id
FROM address
WHERE postal_code IS NULL OR postal_code = '';
```

2. Show cities where the number of residents exceeds a two-digit threshold.

```sql
SELECT ci.city, COUNT(c.customer_id) AS customer_count
FROM customer c
JOIN address a ON c.address_id = a.address_id
JOIN city ci ON a.city_id = ci.city_id
GROUP BY ci.city
HAVING COUNT(c.customer_id) > 99;
```

3. List the titles of films that have never been rented by any customer.

```sql
SELECT f.title
FROM film f
WHERE f.film_id NOT IN (
    SELECT DISTINCT i.film_id
    FROM inventory i
    JOIN rental r ON i.inventory_id = r.inventory_id
);
```

4. Find customers who currently have at least one active rental (i.e., return_date is NULL).

```sql
SELECT DISTINCT c.first_name, c.last_name
FROM customer c
JOIN rental r ON c.customer_id = r.customer_id
WHERE r.return_date IS NULL;
```

5. List addresses that are not associated with any customer.

```sql
SELECT a.address_id, a.address
FROM address a
LEFT JOIN customer c ON a.address_id = c.address_id
WHERE c.customer_id IS NULL;
```

6. Show the first and last names of customers who live in the city `London`.

```sql
SELECT c.first_name, c.last_name
FROM customer c
JOIN address a ON c.address_id = a.address_id
JOIN city ci ON a.city_id = ci.city_id
WHERE ci.city = 'London';
```

7. Display rental IDs and rental dates for rentals made after 1 January 2006.

```sql
SELECT rental_id, rental_date
FROM rental
WHERE rental_date > '2006-01-01';
```

8. Find the rental(s) that had the maximum rental duration (return date minus rental date) in days.

```sql
SELECT rental_id, return_date - rental_date AS duration_days
FROM rental
WHERE return_date IS NOT NULL AND return_date - rental_date = (
    SELECT MAX(return_date - rental_date)
    FROM rental
    WHERE return_date IS NOT NULL
);
```

9. List film categories whose total number of rentals exceeds the average rentals per category.

```sql
SELECT c.name, COUNT(*) AS total_rentals
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
JOIN inventory i ON fc.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
GROUP BY c.name
HAVING COUNT(*) > (
    SELECT AVG(rental_count)
    FROM (
        SELECT COUNT(*) AS rental_count
        FROM film_category fc2
        JOIN inventory i2 ON fc2.film_id = i2.film_id
        JOIN rental r2 ON i2.inventory_id = r2.inventory_id
```

```
        GROUP BY fc2.category_id
    ) t
);
```

10. For each film category, display the film that has been rented the most times.

```
SELECT c.name AS category, f.title, COUNT(*) AS rental_count
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
JOIN film f ON fc.film_id = f.film_id
JOIN inventory i ON f.film_id = i.film_id
JOIN rental r ON i.inventory_id = r.inventory_id
GROUP BY c.category_id, c.name, f.film_id, f.title
HAVING COUNT(*) >= ALL (
    SELECT COUNT(*)
    FROM film_category fc2
    JOIN inventory i2 ON fc2.film_id = i2.film_id
    JOIN rental r2 ON i2.inventory_id = r2.inventory_id
    WHERE fc2.category_id = c.category_id
    GROUP BY fc2.film_id
);
```